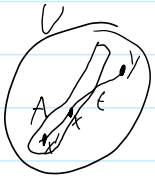
Lecture 10 - 01/07

Monday, July 6, 2026 7:49 PM

we had trapping region U



$\forall t \geq 0 : U(t) = \varphi(t; U) = \bigcup_{x \in U} \varphi(t; x) \in U$

$A = \bigcap_{t \geq} \varphi(t; U)$ which is an invariant set in U

$x \in A \iff \forall t \geq 0 \ \exists y \in U : x = \varphi(t; y)$

to show its invariant: show that $x' = \varphi(t', x)$ for $t' > 0$ and $t' < 0$ ($x \in A$)

$t' \geq 0$:

since $x \in A$, $\forall t \geq 0 \ \exists y \in U : x = \varphi(t; y)$, let $y' = \varphi(t'; y) \in U$

since $x' = \varphi(t'; x)$ and $x = \varphi(t; y)$, $x' = \varphi(t'; \varphi(t; y)) = \varphi(t' + t; y) = \varphi(t; \varphi(t'; y)) = \varphi(t; y')$

so $\forall t \geq 0 \ \exists y' \in U : x' = \varphi(t; y')$

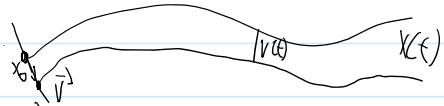
$t' < 0$: $x' = \varphi(t'; x)$,

take t s.t. $t + t' \geq 0$ & $y \in U : \varphi(t + t'; y) = x$

$t + t' = t$

$x' = \varphi(t'; x) = \varphi(t'; \varphi(t + t'; y)) = \varphi(t' + t + t'; y) = \varphi(t; y)$

Lyapunov exponents



$\dot{x} = F(x)$

we want to know if there sensitivity at $x_0 \to \varphi(t; x_0) = x(t)$

take small step from x_0 in direction $\vec{v}$, get to $x_0' \to \varphi(t; x_0') = x'(t)$

$v(t) = |x(t) - x'(t)|$

if $v(t) \xrightarrow[t \to \infty]{} \infty$, x_0 is sensitive

$\dot{v}(t) = \underbrace{DF_{x(t)}}_{A(t)} \cdot \vec{v}$

$A(t)$

if $\|\vec{V}(t)\| \xrightarrow[t\to\infty]{} e^{qt}$

$q(x_0, \vec{v}_0) = \dfrac{\log\|V(t)\|}{t} \quad (t\to\infty)$

simple example

$\dot{X} = F(X)$, FP - X^*

$F(X^*) = 0$

$X = X^* + u$ we can linearize at X^*

$\dot{u} = f(X^*+u) = \underbrace{DF_{(X^*)}}_{A} \cdot u = Au$

A has $AV_1 = \lambda_1 V_1$

$AV_2 = \lambda_2 V_2$

so

$u(0) = c_1 V_1 + c_2 V_2$

$u(t) = c_1 V_1 e^{\lambda_1 t} + c_2 V_2 e^{\lambda_2 t}$, if $\lambda_1 > \lambda_2$, $\|u\| \sim e^{\lambda_1 t}$

so $q(x^*, V_1) = \lambda_1$, $q(x^*, V_2) = \lambda_2$ in general case, there will be only real part of λ

let's say we don't know how to find V_1, λ_1 and V_2, λ_2, what can we do
do the same as before, without knowing these (since it is true for some values)

$u(0) = c_1 V_1 + c_2 V_2$

$u(t) = c_1 V_1 e^{\lambda_1 t} + c_2 V_2 e^{\lambda_2 t}$

look at $\dfrac{\log\|\vec{u}(t)\|}{t}$ and look when $t\to\infty$ this will equal λ_1, the largest $q_1(x^*, \vec{u}_0)$

how do we find λ_2?

take 2 divisions

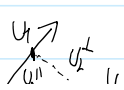
$\vec{u}^1(0) = c_1^1 V_1 + c_2^1 V_2$ $\qquad$ $\vec{u}^2(0) = c_1^2 V_1 + c_2^2 V_2$

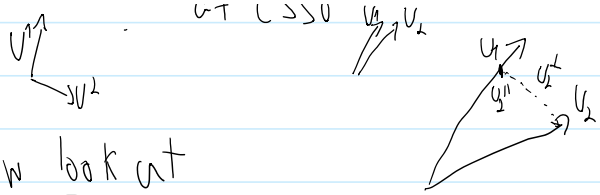
$\vec{u}^1(t) = c_1^1 V_1 e^{\lambda_1 t} + c_2^1 V_2 e^{\lambda_2 t}$ $\qquad$ $\vec{u}^2(t) = c_1^2 V_1 e^{\lambda_1 t} + c_2^2 V_2 e^{\lambda_2 t}$

both are still dominated by λ_1, if you go to large t, they go to it
they will rotate to point in similar direction

at $t=0$ $\qquad$ at $t\gg 0$ V_1, V_2







how look at

do graham schmidt

$$u_2^{\perp} = u_2 - u_1 \cdot \frac{(u_1,u_2)}{\|u_1\|^2} \sim e^{\lambda_2 t}$$

at FP, λ are λ

so $\frac{\log \|u_2^{\perp}\|}{t} \xrightarrow[t\to\infty]{} \lambda_2$

recap, you got $\dot{x} = F(x) \longrightarrow x(t) = \varphi(t, x_0) \longrightarrow A(t)$ (via DF)

$\vec{v_0} \rightsquigarrow \dot{v} = A(t)\vec{v}$

$\|v(t)\| \sim e^{\ell t} \rightsquigarrow \ell_1(x_0, \vec{v_0})$ the largest ℓ

now take $v_0, v_0^{\perp}$ find $v_1^{\perp}$ with graham schmidt,

$\frac{\log \|v_1^{\perp}\|}{t} \xrightarrow[t]{} \ell_2(x_0, t)$

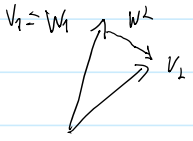
continue to find the rest

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graham schmidt -

you got $\vec{v_1}, \dots \vec{v_n}$, we want to find $\vec{w^1}, \dots, \vec{w^n}$ that are orthogonal

$\vec{w_1} = \vec{v_1}$



how

$$\vec{w^2} = \vec{v^2} - \vec{w_1} \cdot \frac{(\vec{w^1}, \vec{v^2})}{\|w^1\|^2}$$

$$\vec{w^3} = \vec{v^3} - \vec{w_1} \cdot \frac{(\vec{w^1}, \vec{v^3})}{\|w^1\|^2} - \vec{w^2} \cdot \frac{(\vec{w^2}, \vec{v^3})}{\|w^2\|^2}$$

$\vdots$

$\vec{w_n}$

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why are there $\ell_1, \dots, \ell_n$? why not more or less

we have $A(t) = DF(x(t)) \cdot \vec{v}$ and $\dot{\vec{v}} = A(t) \cdot \vec{v}$

the length of v is $\|v\|' = \frac{(v(t), \dot{v}(t))}{\|v\|}$

$$v(t) \sim \underbrace{e^{\int_0^t A(t')\,dt'}}_{B(t)} v(0)$$

$$v(t) = B(t) \cdot v(0)$$

$$\|V^k\|(t) = \left(B(t)\cdot v(0),\ B(t)v(0)\right) = \left(\vec{v}_0,\ B^T(t)B(t)\,v_0\right)$$

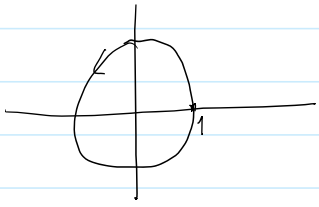
$$\underset{n\times n}{B^T(t)B(t)} \xrightarrow{\ t\ } \Omega^{2t}_{n\times n}$$

so $l_1,\dots,l_n$ are $\lambda_1,\dots,\lambda_n$ of Ω

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example

$$\begin{cases} \dot{x} = -y + x(1-x^2-y^2) \\ \dot{y} = x + y(1-x^2-y^2) \end{cases}$$

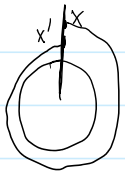


$$\begin{cases} x(t) = r(t)\cos\theta(t) \\ y(t) = r(t)\cdot \sin\theta(t) \end{cases}$$

$$x(t) = \cos t$$
$$y(t) = \sin t$$

$$\dot{\theta} = 1$$
$$\dot{r} = r(1-r^2)$$

What are the Lyapunov exponentials?



we had a poincare map $P(x) = x'$

where if $P'(x) < 1$, its stable,
$P'(x) = 1$ its a orbit
$P'(x) > 1$ its unstable

$P'(x) = 1$ (for $d>2$, this is a matrix and you find its eigenvalues

$r^* = 1$

$$r(0) = 1+u$$
$$r(t) = 1+u(t)$$
$$\dot{u}(t) = (1+u)(-2u)$$
$$\dot{u} = -2u$$
$$u(t) = u_0 e^{-2t}$$
$$l_1 = \frac{\log u(t)}{t} = -2$$
$$l_2 = 0$$